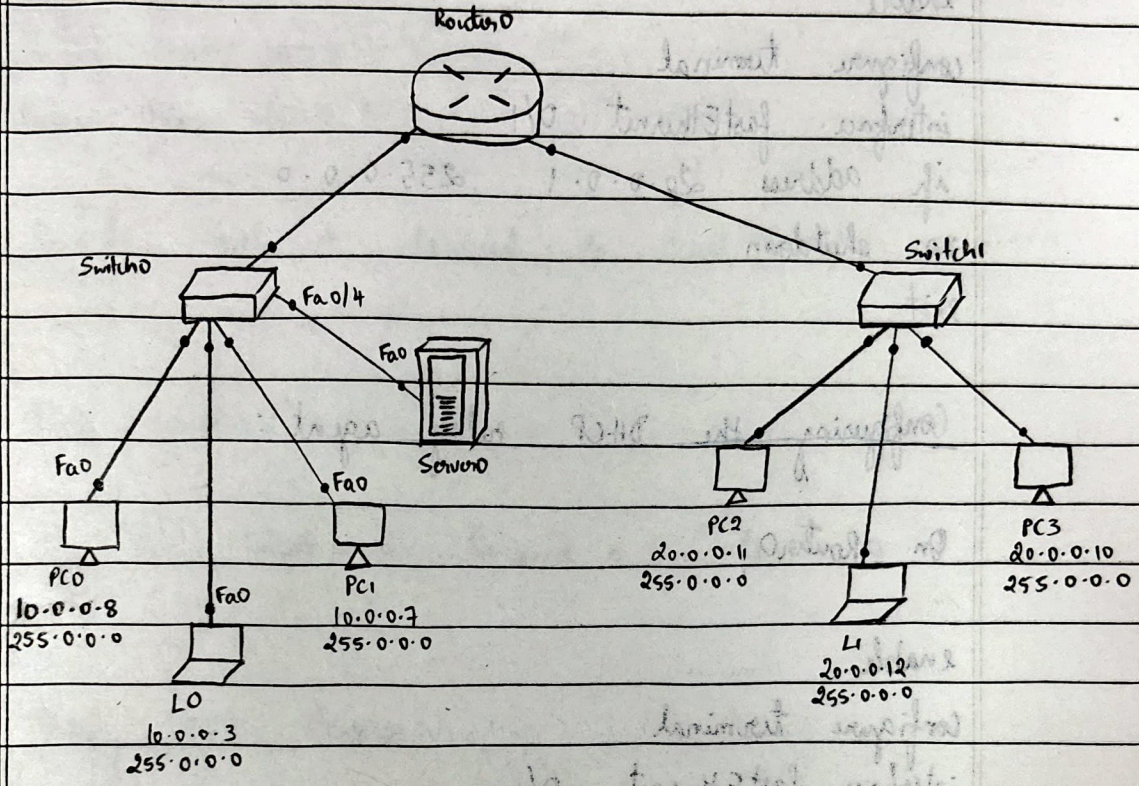
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Lab 3

To configure and verify DHCP for multiple LANs using CPT topology with a router acting as a DHCP relay agent and a centralized server providing IP addresses to clients in different networks.



Add services to DHCP server pool

① BMSCE 1:

Def. Gateway: 10.0.0.1
Start IP Address: 10.0.0.3
Subnet Mask: 255.0.0.0

② BMSCE 2:

Def. Gateway: 20.0.0.1
Start IP Address: 20.0.0.10
Subnet Mask: 255.0.0.0

DHCP Server IP Address: 10.0.0.2

Configuring Router 0:

enable
Configure terminal
interface fastEthernet 0/0

ip address 10.0.0.1 255.0.0.0
no shutdown
exit

enable
configure terminal
interface fastEthernet 0/1
ip address 20.0.0.1 255.0.0.0
no shutdown
exit

Configuring the DHCP relay agent:

On Router0,

enable
configure terminal
interface fastEthernet 0/1
ip helper-address 10.0.0.2
exit
end
write memory

DHCP IP Address configuration of end-devices

Configuration shown in diagram above

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Ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:
Reply from 20.0.0.1 : bytes = 32 time= TTL=
Reply from 20.0.0.1 :
Reply from 20.0.0.1 :
Reply from 20.0.0.1 :

Packets : Sent = 4, Recieved = 4, Lost = 0

Ping 10.0.0.2

Packets : Sent = 4, Recieved = 4, Lost = 0

Verify DHCP Services:

show ip interface brief
show running-config

interface fastEthernet 0/0
ip address 10.0.0.1 255.0.0.0

interface fastEthernet 0/0
ip address 20.0.0.1 255.0.0.0
ip helper-address 10.0.0.2

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Result:

DHCP was configured using the CPT technology. Server0 was configured with 2 DHCP pools for the 10.0.0.0/24 and 20.0.0.0/24 networks. Router0 was configured with ip helper-address to relay DHCP requests from the second LAN to the centralized server. The configuration can be verified using the PC's IP settings, connectivity tests and Packet Tracer Simulation Mode

Lab1 Result:

The IP Addresses were successfully configured on all three routers and both PCs in CPT. Static routing enabled communication between PC0 and PC1, and successful ping replies verified end-to-end connectivity. It also demonstrates Request Timed Out and Destination Unreachable messages under different network failure conditions

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